VINAY SANKAR

GAME DEVELOPER

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Portfolio: terryberry10.github.io/PORTFOLIO-FINAL

PROFESSIONAL SUMMARY

Game Developer with a Bachelor of Computer Applications (BCA) and a post-graduate specialization in Game Technology (M.Sc., in progress). Hands-on with the real-time rendering pipeline through custom HLSL and Shader Graph shaders, particle VFX and GPU-side optimization in Unity (URP), plus prototyping and cinematics in Unreal Engine. Programs in C++ and C# with an object-oriented design approach; familiar with Vulkan, DirectX 11/12 and OpenGL. Experienced in profiling and debugging frame-time problems with the Unity Profiler and Frame Debugger. Based in Bangalore.

TECHNICAL SKILLS & TOOLS

Programming: C++, C#, C, Object-Oriented Design, JavaScript, HTML5

Graphics APIs & Shaders: Vulkan, DirectX 11/12, OpenGL, HLSL, Unity Shader Graph

Game Engines: Unity (Advanced - URP / 2D / 3D), Unreal Engine, Godot

Rendering & VFX: Real-time rendering pipeline, stylized shaders, Shuriken Particles, Combat VFX, Overdraw Control

Profiling & Debugging: Unity Profiler, Frame Debugger, Object Pooling, GC-spike elimination

Animation & Tools: Unity Mecanim, Blend Trees, Ragdoll Physics, Maya / Blender, Git / GitHub

FEATURED GAME PROJECTS

Dottie — 2D Anime-Style 1v1 Combat Game (Unity / C#) — **Solo Developer**

- **GPU Rendering & Custom Shaders:** Engineered stylized HLSL and Shader Graph shaders inside Unity's URP rendering pipeline for dynamic combat auras, afterimage motion trails, and distortion effects inspired by anime fighters.
- **Combat Feel & Game Feedback:** Designed responsive combo attack choreography, special attacks, and punchy audiovisual feedback loops (screen shake, hitstop freeze-frames, hit sparks, physics knockback).
- **Profiling & Performance:** Traced frame-time spikes to garbage collection and fixed them with a modular, object-oriented C# pooling architecture for high-frequency hit particles and projectiles, holding a smooth 60 FPS.
- **Full-Cycle Execution:** Solo developed all gameplay programming, animation states, custom shaders, and particle systems from scratch.

Escape the Miraj & Unreal Cinematics (Unreal Engine) — **Solo Developer**

- **Grapple Hook Tech Demo:** Built a playable grapple-hook movement prototype in Unreal Engine as the foundation for a planned action game.
- **Cinematic Sequences:** Produced two rendered cinematics in Unreal Engine: a car chase and the story intro for Life of an Ant.

Astral Rally — Fast-Paced Multi-Ball Arcade Game (Unity / C#) — **Gameplay Programmer**

- **Shader & Arena Visuals:** Authored dynamic glowing neon shaders and reactive boundary collision materials that respond visually to high-velocity ball impacts.
- **Physics & Mechanics:** Implemented responsive paddle controls, multi-ball trajectory math, collision detection, and custom trail renderers in C#.

Bait — Reverse-Snake Survival Game (Unity / C#) — **Programmer**

- **Visual Indicators & Shaders:** Built custom trail shaders, danger-zone visual cues, and lightweight particle bursts for near-misses and capture events.
- **AI & Movement Logic:** Programmed predator pursuit algorithms and fluid player navigation mechanics in Unity.

FREELANCE EXPERIENCE

Freelance Web & Application Developer — Bangalore, India

- Built and deployed custom web applications and interactive interfaces using JavaScript and HTML5.
- Translated client specifications into clean, modular code with responsive UI and optimized assets.

EDUCATION

M.Sc. in Game Technology • ICAT Design & Media College (Alagappa University) — 2nd Year (Pursuing) • Chennai, India

Bachelor of Computer Applications (BCA) • Lovely Professional University (LPU) — Graduated 2025 • Punjab, India